

ECOV2 Chlorophyll Fluorometer Characterization Sheet

Channel No.: 1

Date: 12/12/2025

S/N: ECOV2-10072

Wavelength: 470 nm

Firmware: 3.10.1

Chlorophyll concentration expressed in $\mu\text{g l}^{-1}$ can be derived using the equation:
CHL ($\mu\text{g chlorophyll l}^{-1}$) = Scale Factor * (Raw Output - Dark counts)

Dark counts, this characterization

Dark counts Range

min: 50 max:

Scale Factor (SF)

Resolution Raw output < 50000 1.15 counts

Raw output $\geq$ 50000 1.00 counts

Standard Error as % of Scale Factor

Specified Range:

min:

Maximum Output (see note below)

Scale Ratio

50 counts

51 counts

5.595E-03 $\mu\text{g CHL l}^{-1} \text{ count}^{-1}$

6.43E-03 $\mu\text{g CHL l}^{-1}$

5.92E-02 $\mu\text{g CHL l}^{-1}$

0.07 %

400 $\mu\text{g CHL l}^{-1}$

3878 $\mu\text{g chlorophyll l}^{-1}$

10.58

- **Raw Output:** Signal output of the instrument. The outputs of the high and low gain stages have been appropriately offset and scaled before combining as follows: If Raw Output < 50000, Raw Output = counts[HI_GAIN]. If Raw Output $\geq$ 50000, Raw Output = (counts[LO_GAIN] - DarkCounts[LO_GAIN]) * ScaleRatio + DarkCounts[HI_GAIN] Note that DarkCounts listed on this sheet are DarkCounts[HI_GAIN]
- **Dark Counts:** The average of 1 minute of collected counts[HI_GAIN] data after 15 minutes of warm-up obtained by covering the detector with the black tape and submersing the sensor in a grounded water bath. The characterization dark counts value is not expected to be the same as the dark counts after sensor integration. Use the dark counts measured after sensor integration for data processing.
- **Dark Counts Range:** Minimum and maximum values recorded in the data collected to determine the dark count value.
- **Scale Factor:** The slope of the linear regression of the target addition series comparing this instrument to a reference instrument.

- **Resolution:** Standard deviation in counts of the data collected to determine the dark counts presented as counts and engineering units by multiplying by the scale factor. For standard deviations of less than one count, resolution is the same as the scale factor.
If Raw Output ≥ 50000 , then resolution should be scaled by Scale Ratio (Resolution * ScaleRatio)
- **Standard Error as % of Slope:** $100 * (\text{Standard error of slope from linear regression}) / (\text{Scale Factor})$. This value represents uncertainty of this calibration relative to that of the reference sensor. It does not represent total uncertainty in the measurement.
- **Specified Range:** Calibrated range of operation
- **Maximum Output:** Maximum potential scaled output of the instrument. This value is generally greater than the specified range.
- **ScaleRatio:** Amplification ratio of high:low gain channels = $(\text{counts}[\text{HI_GAIN}] - \text{DarkCounts}[\text{HI_GAIN}]) / (\text{counts}[\text{LO_GAIN}] - \text{DarkCounts}[\text{LO_GAIN}])$. When converting Raw Output to engineering output, this value is not needed; low gain raw data has already been appropriately offset and scaled.

Characterization Methodology:

We perform a multipoint linear regression analysis of the output of this instrument to the output of a calibrated reference instrument using an additive sequence of increasing loadings of uranine dye. The goodness of fit is determined by the standard error which must be less than 1% of the scale factor. The reference units were calibrated to extracted chlorophyll by measuring the fluorescence response in a monoculture of *Thalassiosira weissflogii* in controlled light conditions and in exponential phase growth. The chlorophyll concentration from the cultures was determined by the extraction and fluorescence method. This establishes an instrument response to chlorophyll concentration scale factor for the reference unit, with respect to the particular culture used. Caution should be observed in using the calibrated value without reference to grab samples.

Note on Maximum Output:

The maximum number of counts that the instrument can output is approximately 160,000. In most natural waters the potential range of the instrument is greater than the measurable range due to interference factors, e.g. the absorbance length scale. Caution should be used for data sets in which the reported values exceed the specified range. Grab samples should be used to confirm values above the specified range.

ECOV2 Scattering Sensor Calibration Sheet

Channel No.: 2
Date: 12/12/2025 S/N: ECOV2-10072
WaveLength: 700 nm Firmware: 3.10.1

$$\beta(\theta_c, \lambda) \text{ m}^{-1} \text{ sr}^{-1} = \text{Scale Factor} \times (\text{Output} - \text{Dark Counts})$$

$$\text{BBP}(\lambda) \text{ m}^{-1} = 2 * \pi * \sin^2 \theta_c [\beta(\theta_c, \lambda) - \beta_{\text{sw}}(\theta_c, \lambda)]$$

Dark counts, this characterization			51 counts
Dark counts Range	min:	50	max: 52 counts
Scale Factor (SF)			1.558E-06 m ⁻¹ sr ⁻¹ count ⁻¹
Resolution	Raw output < 50000	1.15 counts	1.79E-06 m ⁻¹ sr ⁻¹ count ⁻¹
	Raw output ≥ 50000	1.00 counts	1.65E-05 m ⁻¹ sr ⁻¹ count ⁻¹
Standard Error as % of Scale Factor			0.11 %
Specified Range:		min:	0.7 m ⁻¹ sr ⁻¹
Maximum Output (see note below)			1.083E+00 m ⁻¹ sr ⁻¹
Scale Ratio			10.60
Centroid Angle θ_c			124 degrees
Chi factor χ			1.076 sr

- Raw Output:** Signal output of the instrument. The outputs of the high and low gain stages have been appropriately offset and scaled before combining as follows: If Raw Output < 50000, Raw Output = counts[HI_GAIN]. If Raw Output ≥ 50000, Raw Output = (counts[LO_GAIN] - DarkCounts[LO_GAIN]) * ScaleRatio + DarkCounts[HI_GAIN] Note that DarkCounts listed on this sheet are DarkCounts[HI_GAIN].

- Dark Counts:** The average of 1 minute of collected counts[HI_GAIN] data after 15 minutes of warm-up obtained by covering the detector with the black tape and submersing the sensor in a grounded water bath. The characterization dark counts value is not expected to be the same as the dark counts after sensor integration. Use the dark counts measured after sensor integration for data processing.

- Dark Counts Range:** Minimum and maximum values recorded in the data collected to determine the dark count value.

- **Scale Factor:** The slope of the linear regression of the target addition series comparing this instrument to a reference instrument.
- **Resolution:** Standard deviation in counts of the data collected to determine the dark counts presented as counts and engineering units by multiplying by the scale factor. For standard deviations of less than one count, resolution is the same as the scale factor. When Raw Output ≥ 50000 , then resolution is scaled by Scale Ratio (Resolution * ScaleRatio)
- **Standard Error as % of Slope:** $100 * (\text{Standard error of slope from linear regression}) / (\text{Scale Factor})$. This value represents uncertainty of this calibration relative to that of the reference sensor. It does not represent total uncertainty in the measurement.
- **Specified Range:** Calibrated range of operation
- **Maximum Output:** Maximum potential scaled output of the instrument. This value is generally greater than the specified range.
- **ScaleRatio:** Amplification ratio of high:low gain channels = $(\text{counts}[\text{HI_GAIN}] - \text{DarkCounts}[\text{HI_GAIN}]) / (\text{counts}[\text{LO_GAIN}] - \text{DarkCounts}[\text{HI_GAIN}])$. When converting Raw Output to engineering output, this value is not needed; low gain raw data has already been appropriately offset and scaled.
- **Centroid angle θ_c :** The average of scattering of this sensor, weighted by the sensor's angular response function, which determined by sensor and LED illumination geometry.
- **Chi factor χ :** The scalar conversion factor from particulate scattering at the centroid angle θ_c to that of the total particulate backscattering in the backward hemisphere. The χ factors are based on both modeled and/or measured VSF shape analysis in the backward direction (Sullivan & Twardowski, 2009)
- **$\beta_{sw}(\theta_c, \lambda)$** is the contribution to the VSF by the pure seawater at wavelength λ nm defined in equation (2), which depends on temperature and salinity and which is estimated from Zhang et al. (2009). The Matlab code to calculate β_{sw} at the centroid angle θ_c and different wavelengths is available (<http://doi.org/10.17882/42916>).

Characterization Methodology:

We perform a multipoint linear regression analysis of the output of this instrument to the output of a calibrated reference instrument using an additive sequence of increasing loadings of polydisperse spheres. The goodness of fit is determined by the standard error which must be less than 1% of the scale factor. The reference units were calibrated performing a multipoint linear regression analysis of the output of the reference instrument to the calibrated output of scattering from ac-s using an additive sequence of NIST-traceable non-absorbing 1.999 μm beads, where ac-s attenuation is interpreted as scattering ($b=c$) and the scattering phase function β/b has previously been tabulated using Mie theory and the sensor's angular response function. From the reference value for $\beta = (\beta/b)_{\text{table}} * b_{\text{ac-s}}$ and reference sensor counts, a NIST-traceable value for the reference sensor scale factor is determined.

Note on Maximum Output:

The maximum number of counts that the instrument can output is approximately 650,000. In most natural waters the potential range of the instrument is greater than the measurable range due to interference factors, e.g. the absorbance length scale. Caution should be used for data sets in which the reported values exceed the specified range. Grab samples should be used to confirm values above the specified range.

ECOV2 NTU Characterization Sheet

Channel No.: 2

Date: 12/12/2025

S/N: ECOV2-10072

Wavelength: 700 nm

Firmware: 3.10.1

Nephelometric Turbidity Unit (NTU) Scale Factor

NetSignal is defined as (Raw Output - Dark Counts). Turbidity units expressed in NTU can be derived using the equation:

$$\text{NTU} = \text{NetSignal} * X1 + \text{NetSignal}^2 * X2$$

Dark counts, this characterization

Dark counts Range

min: 50 max:

51 counts

52 counts

X1

6.300E-04 NTU count⁻¹

X2

1.230E-11 NTU count⁻¹

Resolution Raw output < 50000

1.15 counts

7.24E-04 NTU count⁻¹

Raw output ≥ 50000

1.00 counts

6.68E-03 NTU count⁻¹

Standard Error as % of Scale Factor

0.48 %

Specified Range:

min:

300 NTU count⁻¹

Maximum Output (see note below)

444 NTU

Scale Ratio

10.60

- **Raw Output:** Signal output of the instrument. The outputs of the high and low gain stages have been appropriately offset and scaled before combining as follows: If Raw Output < 50000, Raw Output = counts[HI_GAIN]. If Raw Output ≥ 50000, Raw Output = (counts[LO_GAIN]-DarkCounts[LO_GAIN])*ScaleRatio + DarkCounts[HI_GAIN] Note that DarkCounts listed on this sheet are DarkCounts[HI_GAIN]
- **Dark Counts:** The average of 1 minute of collected counts[HI_GAIN] data after 15 minutes of warm-up obtained by covering the detector with the black tape and submersing the sensor in a grounded water bath. The characterization dark counts value is not expected to be the same as the dark counts after sensor integration. Use the dark counts measured after sensor integration for data processing.
- **Dark Counts Range:** Minimum and maximum values recorded in the data collected to determine the dark count value.
- **X1, X2:** The coefficients of an orthogonal distance quadratic regression of the target addition series comparing this instrument to a reference instrument.

- **Resolution:** Using NetSignal equal to the standard deviation in DarkCounts[HI_GAIN], the resolution is then estimated using the calibration equation. When Raw Output ≥ 50000 , then NetSignal = standard deviation (typically 1 count) is scaled by Scale Ratio, and again resolution is estimated using the calibration equation. Thus, when Raw Output ≥ 50000 , resolution is scaled by approximately Scale Ratio. □
- **RMS Error as % of Calibrated Range:** $100 * (\text{RMS Error}) / \text{Specified Range}$, where RMS Error determined by regression residuals. This value represents uncertainty of this calibration relative to that of the reference sensor. It does not represent total uncertainty in the measurement.
- **Specified Range:** Calibrated range of operation
- **Maximum Output:** Maximum potential scaled output of the instrument. This value is generally greater than the specified range.
- **ScaleRatio:** Amplification ratio of high:low gain channels = $(\text{counts[HI_GAIN]} - \text{DarkCounts[HI_GAIN]}) / (\text{counts[LO_GAIN]} - \text{DarkCounts[LO_GAIN]})$. When converting Raw Output to engineering output, this value is not needed; low gain raw data has already been appropriately offset and scaled.

Characterization Methodology:

We perform a multipoint linear regression analysis of the output of this instrument to the output of a calibrated reference instrument using an additive sequence of increasing loadings of formazin. The goodness of fit is determined by the standard error which must be less than 1% of the scale factor. The reference units were calibrated to NTU by measuring the response to formazin using a Hach TL2350 Turbidimeter. This establishes an instrument response to NTU scale factor for the reference unit, with respect to formazin. Caution should be observed in using the calibrated value without reference to grab samples.

Note on Maximum Output:

The maximum number of counts that the instrument can output is approximately $65535 * \text{ScaleRatio}$. In most natural waters the potential range of the instrument is greater than the measurable range due to interference factors, e.g. the attenuation length scale. Caution should be used for data sets in which the reported values exceed the specified range. Grab samples should be used to confirm values above the specified range.